

MID-SEMESTER EXAMINATION, SEPTEMBER 2023

Course Code: ECECC08/EIECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 1:30 Hours

Max. Marks: 15

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	A combinational circuit with three inputs, x, y, and z, and three outputs, A, B, and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When The binary input is 4, 5, 6, or 7, the binary output is two less than the input. (a) Draw the truth table. (b) Implement by using NAND gates only	1+ 2	CO1
2	The following Boolean function $F(A, B, C, D) = \Sigma(1, 2, 5, 7, 8, 10, 11, 13, 15)$ (a) Draw the truth table (b) Implement it by using 4X1 MUX and external gates.	0.5+ 2	CO3
3	(a) Convert 110011 from gray code to binary. (b) Perform subtraction using 2's complement 11000.001 - 110.11	1+ 2	CO1
4	(a) Draw the circuit of CMOS two input NAND gate. (b) Represent the function in Canonical standard Sum of Product form. $F(w, x, y, z) = (w + x + y)(\bar{w} + xz) + \bar{y}\bar{z}$	1.5+ 1.5	CO2 CO1
5	Design a BCD to Excess 3 decimal code convertor. (a) Make a truth table (b) Implement it by using a decoder of suitable size and minimal number of inputs to gates.	1+ 2	CO3